

SHIVAM PANDEY

AI Research Engineer – World Models / VLA / ML Systems

HomePage shivampr21.github.io

LinkedIn [shivampr21](https://www.linkedin.com/in/shivampr21)

GitHub [shivampr21](https://github.com/shivampr21)

UNI B.Tech. & M.Tech. @ IITK

Email pandeyshivam2017robotics@gmail.com

Phone +91-7974326386

BLOG shivampr21.github.io/kernelized



AI Research Engineer spanning **embodied intelligence**, **foundation-model research** and **GPU-scale systems**. I work across **VLA development**, **adaptation**, **evaluation and interpretability**, latent world-action modelling, **multi-view reasoning** and **behavior-driven scenario discovery** for autonomous driving, with prior research in long-context architectures and distributed LLM training. My work spans **multimodal representation learning**, **VLM adaptation**, **retrieval-backed reasoning** and **data-centric model improvement** for large-scale perception and decision-making. On the systems side, I build **CUDA/CUTLASS kernels**, **RDMA/GPUDirect data paths** and **low-latency multi-GPU inference stacks**, bridging model research with efficient training, evaluation and deployment.

EXPERIENCE

Senior Engineer / AI Research – Foundation Models for Autonomous Driving

Robert Bosch

Bengaluru, India, Jan'26–Present

World Models

VLA

World-Action Models

Scenario Mining

Multi-View Reasoning

- Led **VLA holdout and targeted data-infusion studies** to quantify generalization under behavior-starved scenarios; demonstrated that only **~6% targeted additional data** improved trajectory **ADE/FDE by ~30%**, motivating scenario-aware sampling over uniform data scaling.
- Developed a scalable **JEPA-inspired VLA diagnostic framework** that projects decision-relevant representations into an event-level behavioral memory, combining latent retrieval, trajectory descriptors and actor-interaction features to diagnose predictions and failure regimes.
- Designed the framework toward **latent world-action modelling**, using action-conditioned representation prediction to compare candidate futures and identify sparse, high-error behavioral regions for targeted scenario mining and data infusion.
- Investigated what constitutes a **faithful Chain of Causality** in autonomous driving, distinguishing genuine causal interaction structure from temporally ordered ego actions and formalizing the required actors, triggers, relations, contextual constraints and maneuver phases.
- Built a **multi-view event-definition and classification pipeline** combining ego dynamics, actor trajectories, lane-stream topology and interaction pressure to detect maneuver onset, escalation, commitment and resolution across merges, crossings, turns, cut-ins and obstacle interactions. (**Patent in Progress**)
- Developed large-scale **rare-event and multi-view scenario discovery** using compositional behavior taxonomy, topology-grounded interaction mining and representation-based reranking; generated **241 scenario windows**, **1.65M assignments across 237K clips** and **2.57M model-safe keyframe candidates** for VLA evaluation and resampling for training.

Senior Computer Vision Engineer

Quidich Innovation Labs

Mumbai, India, Apr'24–Jan'26

Promoted from Computer Vision Engineer to Senior Computer Vision Engineer in Apr'25

- Led development of **multimodal foundation-model systems**, fine-tuning **Qwen3-VL / SmolVLM** with **SFT, QLoRA and DPO** for structured video understanding, and integrating embedding-driven retrieval and agentic reasoning over long-horizon broadcast data.
- Developed a history-aware **diffusion + attention video model** for long-horizon scene understanding, improving temporal consistency through rapid scene transitions while reducing training and inference compute.
- Architected a **GPU-native inference and streaming stack** for LLM/VLM and vision workloads, combining **UCX/UCC, MPI, TensorRT and GPUDirect RDMA** with zero-copy, lock-free data movement between SDI/ConnectX devices, GPU memory and distributed model processes.
- Built fault-tolerant **distributed GPU inference** with process-level redundancy and automated failover, while optimizing model deployment through **PTQ/AWQ, AIMET and TensorRT** for latency- and memory-constrained live systems.
- Delivered real-time **multi-camera perception and tracking** spanning Re-ID, 3D pose, plane tracking and sensor fusion at ≥ 300 FPS, using **DINOv2 embeddings**, GPU optical flow and optimized relocalization; also stabilized DETR training for low-latency sports perception.

AI Research Engineer

Manifest AI

New York, US, Feb'24–Apr'24

- Engineered a **highly parallel training stack** for foundation models, including large language models and linear-attention transformers, leveraging **CUDA and Triton kernels** to **bypass JAX/XLA compilation bottlenecks**.
- Designed and implemented a **sharded multi-node matrix multiplication operator** with compute and communication overlap, effectively **bypassing limitations in JAX/XLA compilation** and improving training throughput.
- Scaled **extensive context-length experiments** for LLMs on **4x8 NVIDIA H100 GPUs** using sophisticated data and model sharding strategies. Implemented **compute-communication overlap with NCCL, MPI** to significantly reduce synchronization stalls.

- **Optimized linear attention for long-context models by implementing state-size scaling through embedding partitioning**, achieving performance on par with standard exponential attention for long-contexts.
- **Delivered compute vs. context scaling laws for linear-attention transformers**, achieving successful training on context windows up to **20K tokens**. Architected the data pipeline to process and filter the **30T-token RedPajama-v2 dataset**, orchestrating distributed processing jobs on GCP with dsub.

Research Engineer Intern @ Five AI

Edinburgh, UK, Aug'22–Oct'22

- Implemented a GNN-based trajectory prediction model for autonomous driving, improving training efficiency via controlled ablation studies and optimization of message-passing architectures.

Software Engineering Intern @ Bosch Global Software Technologies

Bengaluru, India, Jun'21–Jan'22

- Integrated CARLA with ROS2 and a SIL framework (Cloe) to enable automated, CI-driven regression test pipelines.

EDUCATION AND ACHIEVEMENTS

B.Tech & M.Tech

IIT Kanpur

CPI 10.0/10

Thesis: 3D Multi-Modal Multi-Object Tracking with focus on Efficient/Robust Discriminative Manifold Learning, Optimization

JEE Advanced

AIR 3315 (General)

English Proficiency Test (ITEP)

CEFR Level C1

SELECTED COURSES & CERTIFICATES

- Sequence Models
- Reinforcement Learning Specialization
- Machine Learning for Signal Processing
- Algorithmic Toolbox and Data Structures
- Machine Processing of Remotely Sensed Data
- Self-Driving Cars Specialization

INTERESTS

AI Interpretability

VLA

World Models

Compute Scaling

Embodied AI

Reinforcement Learning

Self-Driving Cars

TECHNICAL SKILLS

VLA & Embodied Driving:

VLA

World Models

Vision-Language Models

Trajectory Prediction & Planning

Scenario Mining

Multi-View Perception

Behavioral Evaluation

Representation-Based Interpretability

Imitation Learning

RL

Driving / VLA Models:

Qwen-Drive-1.0

SimLingo

DiffusionDrive

InternVL

Qwen3-VL

SmoIVLM

V-JEPA

DINOv2

World Models & Simulation:

NVIDIA Cosmos

Cosmos Predict

VISTA

ReSim

CARLA

Foundation Models & Training:

Transformers

Pretraining

SFT

PEFT

LoRA / QLoRA

VLM Adaptation

RAG

Quantization (PTQ, AWQ, SmoothQuant)

RL & Post-Training:

RLHF

SFT

DPO

PPO

GRPO

Reward Modeling

Preference Optimization

Policy Optimization

ML Frameworks & Inference:

PyTorch

Hugging Face

DeepSpeed

vLLM

TensorRT

AIMET

JAX / XLA

GPU Programming:

CUDA

Triton

CUTLASS

CuTe

Nsight Systems / Compute

Distributed GPU Systems:

NCCL

MPI

UCX

UCC

RDMA / RoCEv2

GPUDirect RDMA

GDS

GDRCopy

Languages & Infrastructure:

Python

C++

Rust

Docker

Slurm

Kubernetes

GCP

AWS

Azure

Retrieval & Data:

FAISS

Qdrant

Zarr

RESEARCH PROJECTS

- **Traj-JEPA – Causal Representation Analysis for VLA:** Developed a trajectory-conditioned JEPA representation model combining video, ego/actor motion and lane topology, and used the resulting latent space as an interpretability basis for **VLA policies**. Mapped layer-wise VLA activations into physically grounded event representations and performed counterfactual activation patching and policy-sensitivity analysis to trace how interaction concepts propagate into trajectory decisions.
- **Kernelized – Compiler-Oriented Parallelism Analysis:** Developed a mathematical framework for deriving execution parallelism from tensor programs using **data-dependency graphs**, **loop-carried dependency analysis** and **SCC decomposition**, identifying legal loop transformations, tiling and micro-kernel decompositions for operators including Softmax and FlashAttention.

PUBLICATIONS/WRITE-UPS

1. **Patent: Scenario mining for VLA (In Progress)**
2. RMS-ICP: Robust Multi-Scale ICP – improved alignment stability and convergence on noisy, large-baseline scans; applicable to multi-sensor fusion.(Paper Link).
3. Contrastive Learning & 3D MOT – contrastive objectives for robust association under occlusion; fewer ID switches in multi-modal tracking. (Paper Link).
4. 3D Multi-Modal MOT – geometry-aware priors for long-horizon association in sparse-sensor settings. (MS Thesis Link)

POSITION OF RESPONSIBILITY

Teaching Assistant: IITK

Computer Vision, AI, Path & Motion Planning, Sensor Navigation, Geoinformatics

Event Coordinator

ISSTF Open House, IITK